UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

VIETNAM FRANCE UNIVERSITY

INTERNSHIP REPORT

EduFlow – Study Time Management System

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Intake: 2022–2026

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Chapter 1

Introduction

1.1 Context

Life for a student is very busy and stressful. Every day, we go to class, listen to lectures, and go back home with a lot of homework. When we open our notebooks and see five different assignments for the week, we often feel tired and overwhelmed. We do not know what to do first. Because we feel scared of the big pile of work, we often close the book and start playing video games or watching movies instead.

We try to use tools to help us. For example, many students try to use Microsoft To Do, Notion, or other task apps. But these apps are usually made for office workers. They have too many menus, tags, and complex settings. A student just wants a simple place to write "Math homework" and do it. It takes too much time to learn how to use these complex apps.

Also, students like to study with friends to feel less lonely. During the weekend, we often use Discord or Facebook Messenger to call friends and study together. But this is a bad idea. Discord has music bots and memes. Facebook shows us new videos and messages from other people. In the end, nobody studies. We just talk and play for three hours.

Because of all these problems, I decided to build EduFlow. EduFlow is a very simple and clean website. It is made only for students. There are no games, no news, and no distractions. It is just a place to track homework, focus on studying, and learn with friends safely.

1.2 Motivation

I started this project for a few reasons. I wanted to fix my own bad habits, and I wanted to practice my coding skills for my future job.

First, I always have a big problem with procrastination. For example, when a teacher tells me to "Write a 20-page final report", I just stare at the blank screen. I do not know what to write first. So, I do nothing. I wanted my website to fix this using Artificial Intelligence (AI). In my app, if a user types "20-page report", they can click an AI button. The AI will quickly cut this big task into small steps, like "Step 1: Write an outline", "Step 2: Read 3 books". When I see Step 1, it looks very easy, so I can start working right away. No more fear.

Second, I have a problem with forgetting things. I study a math formula on Monday, but when the test comes on Friday, I forget it completely. I read about something called the SM-2 algorithm. It is a simple math rule. It counts the days and tells you exactly when you need to review a lesson before your brain forgets it. I really wanted to write the code for this rule and put it in my website to help me remember better.

Third, studying alone in a quiet bedroom makes me sleepy. I wanted my app to have "Study Rooms". It is like going to a virtual library. You can join a room and see your friends online. You can chat with them. You also see a Pomodoro clock running on the screen. You study hard for 25 minutes, and then the clock tells everyone to rest and chat for 5 minutes. This makes studying feel like a fun game and keeps everyone awake.

Finally, I have a technical reason. I am an ICT student, and I want to be a good web developer after I graduate. I wanted to build a real website from zero to practice. I heard that Python and FastAPI are very fast, so I chose them. I wanted to learn React because it makes websites look beautiful and modern. The hardest part for me was learning WebSockets. I wanted to use WebSockets so that when my friend sends a chat message, it appears on my screen instantly. I do not want to press the F5 key to reload the page.

1.3 Objectives

My main goal for this internship is to build a real, working website. It must run smoothly so real students can use it every day without errors. To do this, I made a list of specific goals.

Functional Objectives:

- **Smart Task Management:** Build a simple page to add tasks. Add an AI button that automatically cuts big, scary tasks into small, easy steps.
- **Review System (SM-2):** Build a system that calculates and saves the review date for each task. It must remind the user when it is time to study again.
- **Online Study Rooms:** Build a place where users can create rooms. Other users can join the room. They can type messages fast. The Pomodoro timer must be exactly the same for everyone in the room.
- **Admin Dashboard:** Build a secret page only for me (the admin). On this page, I can check if the server is running too hot (CPU and RAM usage). I can see how many users are registered, and I can click a button to delete old, useless data.

Technical Objectives:

- **Frontend:** Use React and CSS. The website must look nice and work well on both laptops and mobile phones.
- **Backend:** Use Python and FastAPI. The code must be safe and run fast.
- **Database:** Use PostgreSQL. It must save all user passwords securely.
- **Real-time:** Use WebSockets. It must send chat messages from one user to another in less than one second.
- **AI Connection:** Connect the website to the OpenRouter API. This allows the website to use free AI models to read tasks.

Chapter 2

System Analysis

2.1 System Overview

This chapter explains how EduFlow works inside. The system has two main types of users: Students and Admins.

Students use the website to learn. They can add tasks, use the AI helper, and chat in study rooms. Admins use the website to manage the system. They can see server health, how many students registered, and delete old data.

2.2 Features for Students

Students can do the following things on EduFlow:

- Create a new account and log in safely.
- Add, edit, and delete their study tasks.
- Ask the AI to break a big task into small sub-tasks.
- Join online study rooms to meet other students.
- Send text messages to everyone in the room instantly.
- See a timer (Pomodoro) that is synced with everyone in the room.

2.3 Features for Admins

Admins have a special dashboard. They can:

- See the CPU and RAM usage of the server.
- See a real-time list of new users who just joined.
- See the total number of students and active study rooms.
- Click a button to clean up old, useless data in the database.

2.4 UML Diagrams

To understand the system better, I drew some diagrams.

2.4.1 Use Case Diagram

This diagram shows what Students and Admins can do. Students interact with tasks and rooms. Admins only look at the system data.

2.4.2 Sequence Diagram

This diagram shows the step-by-step flow of data. For example, when a user asks AI to split a task, it shows how the frontend sends data to the backend, how the backend calls the AI API, and how the data is saved in PostgreSQL.

Chapter 3

Methodology

3.1 System Architecture

EduFlow uses a modern Client-Server model. It has three main parts:

- **Frontend (Client):** This is what the user sees. It is built with React.
- **Backend (Server):** This handles all the logic and rules. It is built with FastAPI.
- **Database:** This saves all the data. It uses PostgreSQL and Redis.

3.2 Tools I Used

For coding, I used Visual Studio Code. I used Git and GitHub to save my code history. To put the website on the internet, I used my own Raspberry Pi server and Cloudflare Tunnels.

3.3 How I Built The Frontend

I built the user interface with React.js. I used React Hooks to manage data on the screen. To make it look nice, I used Tailwind CSS. Tailwind helps me write CSS very fast. For the chat feature, I used WebSocket in JavaScript to receive messages without reloading the page.

3.4 How I Built The Backend

I chose FastAPI for the backend. It is a very fast Python framework. I used SQLAlchemy to talk to the PostgreSQL database. This keeps the database safe from hackers.

To handle the chat rooms, I wrote a Connection Manager in Python. When a user sends a message, this manager finds all other users in the same room and sends the message to them instantly.

3.5 How I Handled Authentication

To keep user accounts safe, I used JWT (JSON Web Tokens). When a student logs in, the backend checks the password. If it is correct, it gives the student a token. The student uses this token to do everything on the website safely.

3.6 How I Handled the Database

My PostgreSQL database has different tables to store information clearly:

- **Users Table:** Saves email and password.
- **Tasks Table:** Saves the homework name, due date, and AI sub-tasks.
- **Rooms Table:** Saves the study room name and messages.

Chapter 4

Results and Discussions

4.1 Project Results

I successfully finished building EduFlow. It can do everything I planned in Chapter 1.

Task and AI Features: The AI task splitting works very well. When a student types a big task, the backend talks to the OpenRouter API. After a few seconds, the student gets a list of small, easy steps.

Study Rooms and Chat: The study rooms work fast. When a student sends a message, it appears on the other student's screen in less than 1 second. The Pomodoro timer is also the same for everyone in the room.

Admin Dashboard: The admin dashboard successfully shows real data. It connects to the database to count exactly how many students are registered.

4.2 Challenges I Faced

Building this website was hard. I faced two big problems.

First, the free AI API was sometimes very slow or failed to answer. To fix this, I wrote code to automatically switch to a backup AI model if the first one fails. This makes sure the user always gets an answer.

Second, the WebSocket chat had bugs. If a user opened two tabs in their browser, the chat messages would duplicate. To fix this, I had to update my Connection Manager to track each user better and not send the same message twice.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This internship project was a great learning experience. I learned how to build a full web application from start to finish. I learned how to use FastAPI, React, PostgreSQL, and WebSockets together. EduFlow is now a real, working website that helps students study better and stay focused. I am very proud of what I built.

5.2 Future Plans

In the future, I want to add more things to EduFlow:

- **Mobile App:** I want to make an app for iOS and Android so students can use it on their phones.
- **Better Matching:** I want to use Machine Learning to help students find the best study partners, not just by subject.
- **Offline Mode:** I want to let students see their tasks even when they do not have WiFi.